

Unlocking Nighttime Mobility: Land Use and Accessibility in Public Transit for Night Commuters

ABSTRACT

Night commuters are integral to urban transportation systems. Essential services such as healthcare and manufacturing rely on workers who travel at night, and reliable mobility options are crucial for them. A gap exists in understanding how land use and accessibility influence public transportation use among night commuters. This study addresses this gap by using public data to explore land use and accessibility factors that affect night commuters' public transportation use in New York State. We investigated (1) the demographic characteristics of night commuters; (2) the influence of land use and accessibility on nighttime public transportation use; and (3) potential improvements to increase public transportation use and their impact. We combined data from the National Household Travel Survey with the Smart Location Database to link home locations with land use characteristics. Using logistic regression, we found that although females are generally less likely to be night commuters, they are more likely to use public transportation. Longer commute distances are associated with higher use of public transportation. Increasing job density along fixed-guideway transit routes and improving overall job accessibility via public transportation significantly enhances public transportation use among night commuters. This research provides actionable insights for public transportation agencies and urban planners to support night commuters, improving access and encouraging nighttime employment.

Keywords: Night commuters; public transportation; land use; job accessibility; Smart Location Database; National Household Travel Survey

1. INTRODUCTION

Night commuters are essential to the functioning of key sectors such as healthcare, hospitality, and manufacturing, yet their transportation needs are often overlooked in traditional planning models. These models primarily focus on daytime travel patterns, failing to account for the unique challenges faced by night commuters. A 2019 study by the American Public Transportation Association found that late-shift workers relying on public transit experience commute times that are, on average, twice as long as those who drive (1). This disparity highlights a critical gap in transportation planning, where limited nighttime transit options can exacerbate inequities and reduce workforce accessibility. Addressing this issue is not just about convenience—it has significant implications for sustainability, social equity, and environmental impact. By integrating night travel considerations into broader transportation strategies, planners can create a more inclusive and resilient transit system that supports economic mobility while reducing reliance on personal vehicles.

Efforts to enhance nighttime public transportation have included strategies such as implementing new mobility technologies like ride-sharing for flexible services and expanding bus routes for late-night workers, such as King County's Night Owl service in Washington (2–4). Additionally, studies have examined factors influencing night commuters' transportation choices, including concerns about safety, long travel times, and high perceived costs (5, 6).

However, a significant research gap remains in understanding how land use and public transportation accessibility factors influence transit use specifically among night commuters. While extensive research has demonstrated that strategies such as increasing job density along transit corridors can shift daytime commuters from personal vehicles to public transportation (7, 8), it is unclear whether these approaches are equally effective for night commuters. Furthermore, many prior studies rely on survey methods that focus on small, homogeneous regions, limiting their ability to capture the systemic, macro-level factors necessary for broader urban planning and policymaking (9).

To address these gaps, this study employs a data-driven approach using a case study in New York State (NYS). The key research question guiding this study is: How do land use and transportation accessibility influence public transit use among night commuters, and what policy interventions can enhance transit accessibility and ridership? Specifically, it seeks to answer three research questions:

- 1) What are the demographic characteristics of night commuters? Understanding these considerations helps identify unique attributes of night commuters compared with the general population.
- 2) What are the land use and public transportation accessibility factors of existing public transportation users among night commuters? This information helps show the distinct characteristics of current public transportation users compared with non-users.
- 3) What can be improved to promote more public transportation use among night commuters? How would the likelihood of this behavior increase with specific improvements in these factors?

Instead of relying on time- and resource-intensive surveys with often limited sample sizes, this study leverages publicly available datasets. These include the 2017 National Household Travel Survey (NHTS), which provides detailed individual trip data, and the Smart Location Database (SLD), which offers information on land use and public transportation accessibility. By integrating these datasets, the study develops a cost-effective framework for urban planners and transit agencies to strategically identify where night commuters are located as well as who may need or be inclined to use public transportation. This framework enables targeted investments in public transit, ensuring resources are allocated efficiently to align night commuting demand with adequate supply. Ultimately, this approach supports effective decision-making to maximize the impact of transportation service improvements.

2. LITERATURE REVIEW

Understanding the travel characteristics and public transportation accessibility for night commuters is crucial in urban transportation planning and has received increasing attention in recent years. Commuters who travel during off-peak night hours often face distinct challenges compared with daytime commuters, and addressing their mobility needs can significantly enhance their quality of life and contribute to broader sustainability goals. This literature review aims to synthesize existing research on night commuter travel behavior, their public transportation challenges, and the adopted solutions to improve public transportation services during night hours. The findings from this literature review are summarized in Table 1.

Table 1. Summary of findings from the literature

Category	Findings	References
Socioeconomic and travel characteristics of night commuters	- Primarily employed in essential services such as healthcare, hospitality, and manufacturing - Higher proportion of lower-income and minority groups than daytime commuters - Unique travel patterns compared with daytime commuters	(10, 11)
Public transportation challenges and mode choice factors for night commuters	- Reduced service frequency and longer wait times during late-night hours - Lower transit service availability at night, especially in rural areas - Safety concerns regarding heightened sleepiness and impaired driving performance among night workers - High reliance on private vehicles because of unreliable transit services - Increased likelihood of using cars over other modes at night, influenced by factors such as gender, urbanity, and weather	(1, 12–14)
Built environment and land use policies' effect on public transportation use	- Higher urban density, mixed land uses, and proximity to transit stops linked to increased public transportation use - Increased jobs and amenities around transit stops to encourage shifts from private vehicles to public transit	(15–18)

A substantial body of research highlights the unique travel patterns and socioeconomic characteristics of night commuters. Studies indicate that night commuters are often employed in essential services such as healthcare, hospitality, and manufacturing—sectors that operate around the clock and rely heavily on workers during off-peak hours (10, 11). This demographic tends to include a high proportion of low-income and minority groups, underscoring the critical need for equitable and reliable transit services tailored to their needs. Their other activities also differ significantly from that of regular commuters. Studies have shown that night and evening shift workers are less likely to make trips for recreation, visiting others, or dining out on workdays. Similarly, evening shift workers are less inclined to engage in community or religious activities. Additionally, evening shift workers are less likely to run household errands on workdays, with this effect being particularly pronounced among women, especially for tasks such as shopping and other errands (19).

The frequency and reliability of nighttime transit services are major concerns for night commuters. Reduced service frequency and longer wait times during late-night hours have been identified as significant barriers to public transportation use (1). Research suggests that transit service availability is significantly lower at night, especially in rural communities (20, 21). Increasing the frequency and

1 reliability of nighttime transit could enhance travel experiences for night commuters, boost ridership, and
2 reduce reliance on private vehicles. Cities with improved nighttime services have reported higher public
3 transit use and decreased car dependency (13). Discrete choice models revealed that individuals are
4 significantly more likely to use cars over other modes of travel at night, with variations based on gender,
5 urbanity, and weather (12). Safety concerns are critical; night workers often experience heightened
6 sleepiness on their commutes, which impairs driving performance and increases the risk of accidents (14).
7 Addressing these risks through targeted interventions, such as late-night shuttle buses, can improve safety
8 and mobility (22).

9 Built environment factors and land use policies play a crucial role in shaping transit use among night
10 commuters. For general commuters, higher urban density, mixed land uses, and proximity to transit stops
11 are strongly associated with increased public transportation use (15–17). Additionally, creating job
12 opportunities and amenities near transit stops has been shown to encourage shifts from private vehicles to
13 public transit (23). However, it remains unclear whether these relationships hold true for night
14 commuters, as their unique needs and travel patterns have not been thoroughly studied. This knowledge
15 gap calls for more detailed research to understand how specific service enhancements and land use
16 policies can influence night commuters' travel behavior. For instance, enhanced service quality and
17 expanded transit options, such as Bus Rapid Transit and ridesharing, have been effective in boosting
18 transit use among general commuters (18, 24). Yet, further investigation is needed to determine the
19 impact of these measures on night commuters and to identify strategies that effectively address their
20 specific requirements.

21 Current research underscores the importance of addressing the unique mobility needs of night commuters.
22 However, research on the targeted initiatives and service improvements for night commuters' travel is
23 still limited. A thorough quantitative investigation, incorporating land use and socio-demographic
24 characteristics, is needed to understand where and how to improve public transportation services,
25 particularly in terms of land use and public transportation accessibility, for night commuters.

26 Methodologically and data-wise, many prior studies rely on survey methods (11, 25). Regional surveys
27 may provide valuable insights into local nighttime travel behavior but often fail to account for variability
28 across diverse urban and rural contexts or larger-scale macro-level factors. Additionally, as night
29 commuters represent a relatively small proportion of the workforce, analyzing this group requires either a
30 large base sample or specific data collection procedures tailored to nighttime travel patterns. Without such
31 measures, survey results risk being unrepresentative. Consequently, widespread data collection efforts for
32 night commuters are rare, making it difficult for individual researchers to study this population on a larger
33 geographic scale. Some studies have turned to passive mobility data, such as smart card usage records, to
34 address these challenges. While these datasets provide high-resolution travel data, they lack direct
35 information on individuals' socioeconomic characteristics, limiting their utility for comprehensive
36 analyses (26).

37 To overcome these limitations, there has been a growing shift toward leveraging large-scale, publicly
38 available datasets in transportation research. The NHTS, for example, offers granular individual trip data
39 across diverse demographics and geographic scales, supporting analyses on topics such as mode choice
40 and the impacts of vehicle electrification on travel demand (27, 28). Similarly, the SLD provides detailed
41 land use and transportation accessibility information at a fine spatial resolution (Census Block Group
42 level), making it a valuable resource for examining urban form and its effects on mobility (29, 30). By
43 integrating these datasets, this study seeks to expand on the existing body of research while addressing the
44 methodological constraints of smaller, survey-based studies. Specifically, by leveraging both datasets, we
45 explore the combined influence of socioeconomic characteristics and land use factors on nighttime
46 mobility, offering insights that are both scalable and actionable for urban planning and policy
47 development.

3. METHODOLOGY AND DATA

3.1 Definition and Data

In this study, night commuters are defined as individuals who had at least one work trip starting between 6 p.m. and 3 a.m. (Figure 1). This time span was selected to capture a variety of work schedules typically associated with evening, night, and early-morning shifts, as opposed to the traditional 9 a.m. to 5 p.m. workday. For example, trips starting 6 p.m. could reflect the beginning of evening shifts. This definition aligns with standard categorizations in transportation, workforce, and health research, which often use similar time frames to address the needs and behaviors of late-shift workers (31, 32).

Among these night commuters, those who use public transportation for at least one work trip per day are classified as existing public transportation users. The work trip is defined using the variable recorded in the travel survey dataset (detailed in Table 2) that captures the purpose of each trip recorded in the respondent's travel diary. Specifically, this variable represents trips that could be either to or from work. Additionally, the dataset records the mode of transportation used for each trip, including public transportation. Those who did not use public transportation are considered potential public transportation users. This study primarily focuses on understanding how these potential users can become public transportation users.

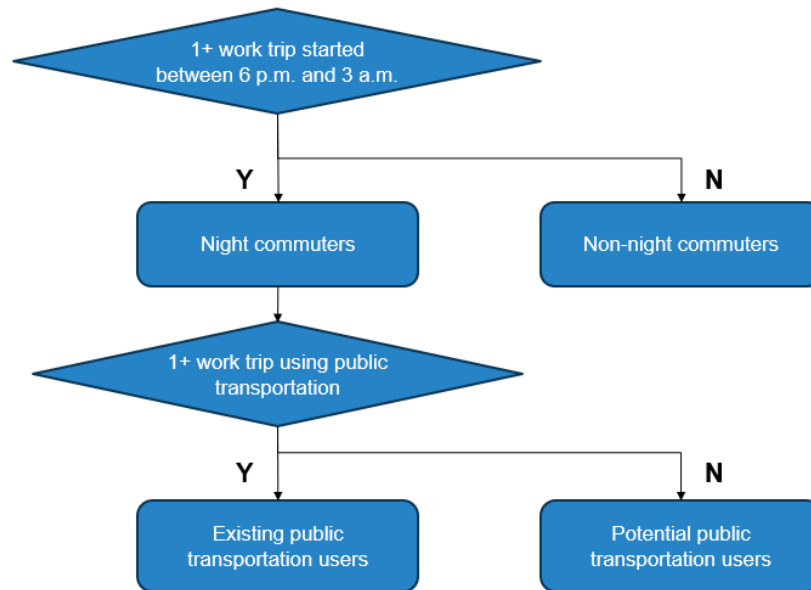


Figure 1. Definition of night commuters and public transportation users.

3.1.1 National Household Travel Survey (NHTS)

One of the primary objectives of this study is to identify the characteristics of individuals likely to be night commuters and, among them, those likely to be existing public transportation users. Previous research has shown that where people live, along with their demographic and socioeconomic characteristics, can influence travel behavior (33, 34). Many studies have used travel surveys, which typically collect daily travel information, including trip frequency, travel distance and time, mode of transportation, and trip purpose (35, 36). We used travel survey data for this study, specifically the 2017 NHTS add-on data, which include household locations and trips made by household members (37). The NHTS, sponsored by the Federal Highway Administration, is a national travel survey of US households. The 2017 NHTS surveyed more than 129,000 households, 26,000 of which from a national sample and the remainder from add-on samples purchased by 13 state or Metropolitan Planning Organization partners.

NYS participated as a partner in the 2017 NHTS, providing access to its add-on data, which included home location information for each survey respondent—data not available in the public version of the national survey. For this analysis, the study region was in NYS, which comprised 34,297 individuals, of whom 3,312 were identified as night commuters. To ensure privacy, the data was aggregated into hexagonal spatial units with a vertex-to-vertex diameter of 3.5 miles.

Using the NHTS add-on data, we analyzed individuals' household location, personal and household attributes, and whether they are classified as night commuters or public transportation users based on the definitions provided earlier. The socioeconomic attributes of individuals and households that might influence their status as night commuters or public transportation users, as adapted for the NHTS data, are summarized in Table 2. Variables were converted into binary format, and for those with multiple categories, one category was chosen as the reference.

Table 2. Description of the socioeconomic variables for all individuals

Attribute	Type	Variable categories	Mean and standard deviation
Home ownership	Household	Rent = 1, own = 0	0.80 (0.40)
Race	Household (primary household respondent)	White = 1, non-white = 0	0.89 (0.31)
Vehicle ownership	Household	0 vehicle, others = 0	0.06 (0.25)
		1 vehicle = 1, others = 0	0.24 (0.43)
		2 vehicles = 1, others = 0	0.42 (0.49)
		3+ vehicles = 1, others = 0	0.27 (0.45)
Annual income	Household	Equal or below \$74,999 = 1, higher than \$74,999 = 0*	0.49 (0.50)
Education	Individual	High school or less = 1, college or higher = 0	0.51 (0.50)
Age	Individual	Between 18 and 24 = 1, others = 0	0.05 (0.23)
		Between 25 and 39 = 1, others = 0	0.15 (0.35)
		Between 40 and 65 = 1, others = 0	0.39 (0.49)
Great circle distance to job locations	Individual	Long distance to work (i.e., over 20 mi) = 1, Not long distance to work = 0	0.063 (0.24)
Gender	Individual	Female = 1, male = 0	0.53 (0.49)
Night commuter	Individual (derived)	Yes = 1, no = 0	0.097 (0.30)
Mode of transportation: Public transportation	Individual trip	“TRPTRANS” variable: public or commuter bus, Amtrak / commuter rail, subway / elevated / light rail / streetcar.	-
Trip purpose: Work	Individual trip	“WHYTRP90” variable: to/from work, worked-related business	-
Existing public transportation user	Individual (derived)	Yes = those who use public transportation for at least one work trip recorded in the travel survey, no = otherwise	0.024 (0.15)

*\$74,999 is the upper limit of an income bracket in the NHTS; the lower limit of this bracket is \$50,000. This upper limit is chosen for categorization because it is the closest to the annual median household income in NYS, which was \$62,756 in 2017 (38).

A limitation of the NHTS definition for existing public transportation users is that it only provides trip data for the specific travel day assigned to each respondent. This may result in including infrequent users as existing users and excluding occasional users who did not use public transit on that day.

3.1.2 Smart Location Database (SLD)

Land use and public transportation accessibility factors are crucial in determining public transportation use and can provide insights into how current land use and transit policies influence this choice (39). By examining these factors for night commuters and their use of public transportation, we gained practical insights into potential improvements.

To analyze the land use and public transportation accessibility factors data, we used the Environmental Protection Agency's SLD, which incorporates data from sources such as the 2018 American Community Survey for vehicle ownership and population, and 2018 HERE Technologies' Maps for geographic features. The SLD provides a comprehensive range of built environment factors (40).

Based on existing literature, we investigated how land use characteristics and transit accessibility affect night commuters and public transportation users. Specifically, we examined the following factors from the SLD:

- Transit frequency during evening peak hours: Because night commuters may leave for work during the evening, high transit frequency during these hours could indicate better accessibility. Although data for late-night transit frequency were unavailable, peak-hour frequency could suggest the popularity of a stop, which might influence evening use.
- Number of jobs near fixed-guideway transit stops: Fixed-guideway systems such as subways and light rail, which often operate from early morning until late at night, can offer convenience for night commuters. Higher job density near these stops can enhance access to employment and support public transportation use.
- Number of jobs within a 90 min transit commute in a Census Block Group (CBG): This attribute reflects how accessible jobs are within a CBG using public transportation. A higher number indicates better overall employment reachability, providing a viable alternative to personal vehicles for commuting. We also examined one index variable that reflects how a CBG compares to others in terms of transit accessibility within a metropolitan statistical area.

Table 3 summarizes the land use and public transportation accessibility attributes from the SLD that may influence night commuting and public transportation use.

Table 3. Land use and public transportation variables in the SLD

Categories	Variables	Mean, standard deviation
Land use	Road network density (mi/mi ²)	11.65, 9.97
	Employment density (jobs/acre)	5.45, 42.44
	Residential density (households/acre)	4.87, 19.41
	Employment and household entropy (incorporates both employment and occupied housing in its calculation to assess the diversity of activity within a CBG)	0.51, 0.20
Public transportation accessibility	Average number of minutes from service during evening peak period within 0.25 mi of CBG boundary (i.e., 4:00 p.m. and 7:00 p.m. local time)	8.50, 31.00
	Number of jobs within 0.25 mi of fixed-guideway transit stop	59.83, 1,124.53

Number of jobs within 0.5 mi of fixed-guideway transit stop	80.62, 1,189.00
Number of jobs within 90 min public transportation travel time (including transit, walking access, walking egress, wait and transfer times) from residence locations (jobs could be outside CBG)	155,195.47, 606,614.09
Proportional public transportation accessibility to jobs (=Number of jobs within 90 min public transportation travel time from CBG/total jobs within 90 min public transportation travel time from CBG within the whole metropolitan statistical area)	0.070, 0.17

NHTS and the SLD provide attributes for every CBG. To align with the individual sample from the NHTS, we matched individuals' data to the corresponding SLD attributes based on the CBG of their household location.

3.2 Model

The logistic regression model was used in this study because of its statistical robustness and suitability for binary outcome analysis. The logit model was used because it is particularly effective in estimating the probability of an event occurring, which in this case involves being a night commuter and, among night commuters, using public transportation. By modeling the log-odds of these outcomes, we quantitatively assessed the influence of various predictors, such as demographic, socioeconomic, and travel behavior variables, on the probability of the target events.

3.2.1 Model for research question 1: Binary logistic regression for night commuting

To answer the first research question, we used a binary logistic regression model to estimate the probability of an individual being a night commuter. The model specification is as follows:

$$\text{Logit}(P(\text{Night Commuters} = 1)) = \beta_0 + \sum_i \beta_i X_i,$$

where

- $\text{Logit}(P(\text{Night Commuters} = 1))$ is the log-odds of being a night commuter,
- X_i are the independent variables in Table 2 and Table 3,
- β_0 is the intercept term, and
- β_i are the coefficients corresponding to each independent variable X_i .

3.2.2 Model for research question 2: Binary logistic regression for night commuting

To answer the second research question, following the same modeling approach, we used a binary logistic regression model to estimate the likelihood of using public transportation among night commuters. The model specification is as follows:

$$\text{Logit}(P(\text{Public Transportation User} = 1 | \text{Night Commuters} = 1)) = \alpha_0 + \sum_i \alpha_i X_i,$$

where

- $\text{Logit}(P(\text{Public Transportation User} = 1 | \text{Night Commuters} = 1))$ is the log-odds of a night commuter being an existing public transportation user,
- X_i are the variables in Table 2 and Table 3,
- α_0 is the intercept term, and
- α_i are the coefficients corresponding to each predictor variable X_i .

3.2.3 Model for research question 3: Sensitivity analysis

To understand how changes in land use and public transportation attributes affect the likelihood of night commuters using public transportation, we conducted a sensitivity analysis based on the estimated logistic regression model. Specifically, the analysis examined the following:

- Land use changes: Changes in the land use and public transportation accessibility factors, such as increased residential density, mixed-use development, and improved pedestrian infrastructure
- Public transportation accessibility changes: Enhancements in public transportation service frequency, coverage, and accessibility

For each night commuter in the dataset who was not currently a public transportation user, we calculated the likelihood of them becoming a user given specific changes in land use or public transportation attributes. The model generated probabilities of public transportation use for night commuters in NYS. Using this information, we identified areas with the highest potential increase in public transportation use under hypothetical improvements in public transportation service or land use.

The sensitivity analysis assumed constant socioeconomic characteristics (e.g., income, education, employment) and focused on how changes in land use and public transportation accessibility individually affect public transportation use. We assumed that individuals with similar demographic profiles would exhibit similar travel behaviors under comparable conditions. This approach helped identify regions where investments in transit infrastructure or land use planning could yield the greatest benefits.

4. STUDY REGION AND DESCRIPTIVE ANALYSES

Figure 2 illustrates the distribution of night commuters' home locations within each hexagon. This spatial representation highlights the areas with significant numbers of night commuters. Figure 2 (a) shows a higher concentration of night commuters in urban areas and major cities across NYS. Figure 2 (b) depicts the proportion of night commuters relative to the total sample size, with shaded areas indicating regions with fewer than five total samples. The analysis reveals that areas without night commuters generally correspond to areas with very few total samples, highlighting the impact of sample size on observed patterns.

We then examined the annual household income distribution between night commuters and other travelers (**Figure 3**). The analysis revealed that night commuters tended to have higher incomes compared to other travelers. A consistently greater share of night commuters fell into the income bracket above \$75K, which exceeds the median household income in NYS of \$62,756 in 2017 (38). However, this finding contrasts with some literature, which often associates night work with lower-income individuals (41). One possible explanation is the study's definition of night commuters, which includes individuals departing from work to home after 6 p.m. This may encompass individuals working overtime, who are not typical shift workers and are often higher-income professionals. As a result, the dataset could include a mix of night commuters with varying roles, such as lower- to middle-income service or logistics workers and higher-income healthcare professionals, which could skew the income distribution. However, the NHTS lacks detailed occupation data to distinguish between these groups. It is also possible that higher-income night commuters were overrepresented in the NYS subset of the NHTS dataset.

Figure 4 compares the mode share for work and non-work trips between night commuters and other travelers, segmented by urban and rural areas (based on the NHTS household location variable). Personal vehicles were the dominant mode across all groups. However, an interesting trend emerged in urban areas: night commuters used public transportation more for work trips than for non-work trips, as well as more than non-night commuters for work trips. This is likely due to the availability of public transportation networks in urban areas, even during off-peak hours. In contrast, rural areas showed a clear reliance on personal vehicles, with over 85% of trips in all categories, and minimal public transportation

use for both night commuters and other travelers, reflecting the limited transit infrastructure in these regions.

Figure 5 shows that the home locations of existing public transportation users were predominantly concentrated in New York City (NYC). This concentration is likely because of the extensive public transportation services available in the city, including fixed-guideway transit systems. The area between Elmira and Ithaca also exhibits a relatively higher concentration of public transportation users. This may be attributed to Ithaca being home to Cornell University. University towns often have a mix of student and service-oriented jobs that require non-standard hours. Bus and rail lines in the area support commuters who live in these regions and travel to Ithaca for work.

Figure 6 and Figure 7 display the number of jobs within 0.25 and 0.5 mi of fixed-guideway transit stops, respectively, according to the SLD dataset. The maps show a significant number of jobs in NYC, extending northward into areas such as Westchester County and Putnam County. Additionally, the city of Buffalo also has some fixed-guideway transit services, contributing to job accessibility in that area.

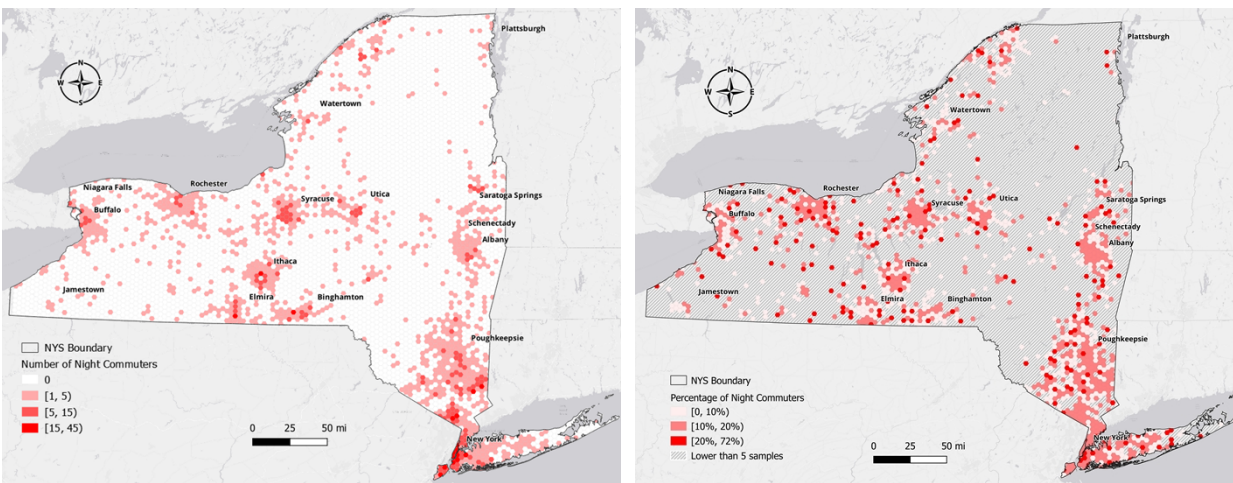


Figure 2. Distribution of night commuters.

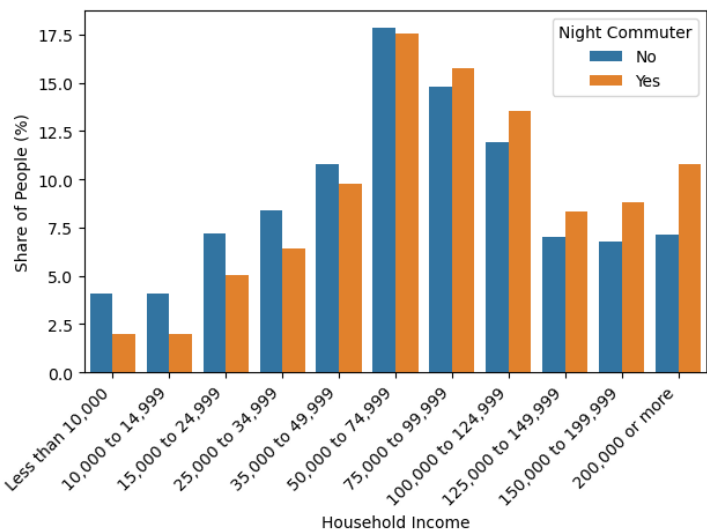
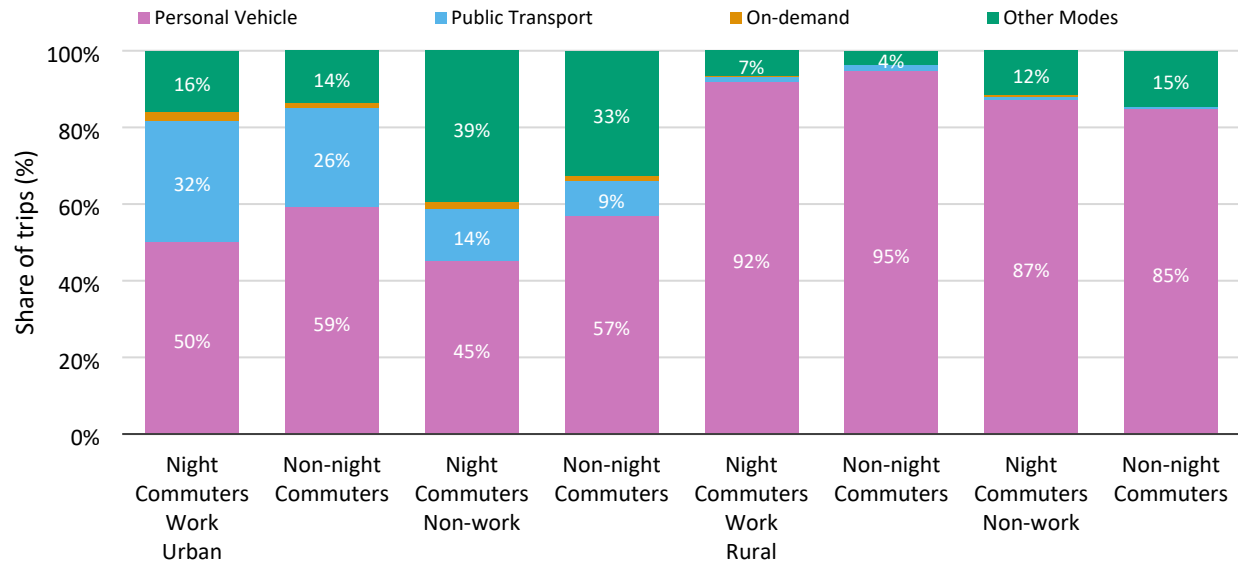


Figure 3. Income distribution between night commuters and non-night commuters.

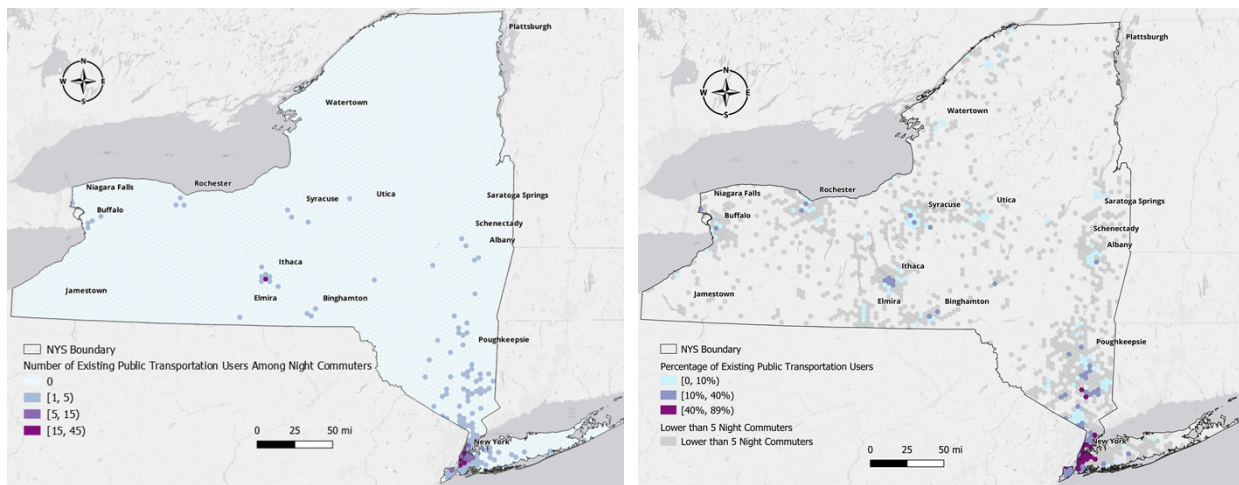
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Figure 4. Mode of transportation used by night commuters and non-night commuters.



(a) Number of existing public transportation users.

(b) Percentage of existing public transportation users among night commuters.

4

Figure 5. Distribution of existing public transportation users among night commuters.

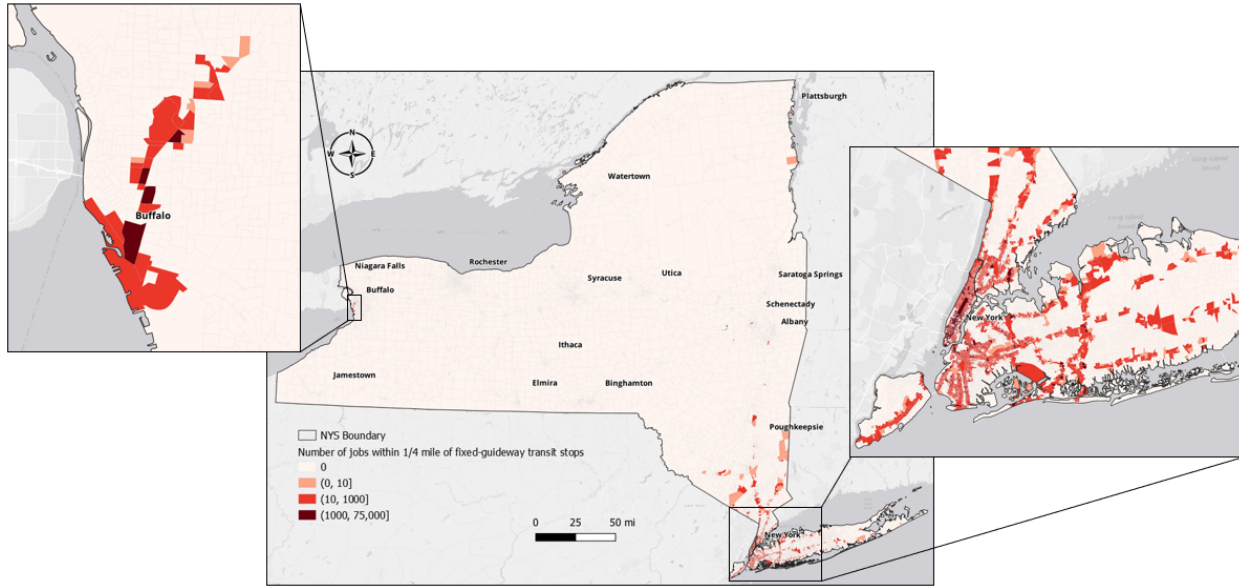


Figure 6. Number of jobs 0.25 mi from a fixed-guideway transit stop.

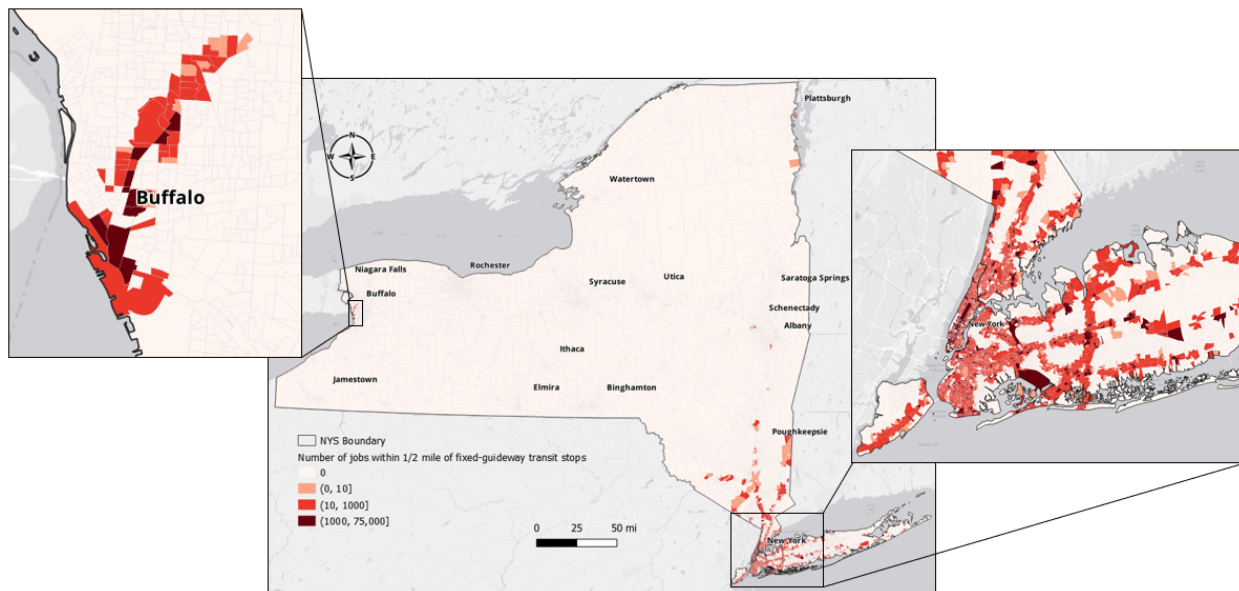


Figure 7. Number of jobs 0.5 mi from a fixed-guideway transit stop.

5 Figure 8 shows the residence locations from which people can reach jobs within a 90 min public
 6 transportation commute. The map reveals that major cities such as Rochester, Buffalo, Syracuse, Albany,
 7 and NYC offer convenient access to jobs via public transportation.

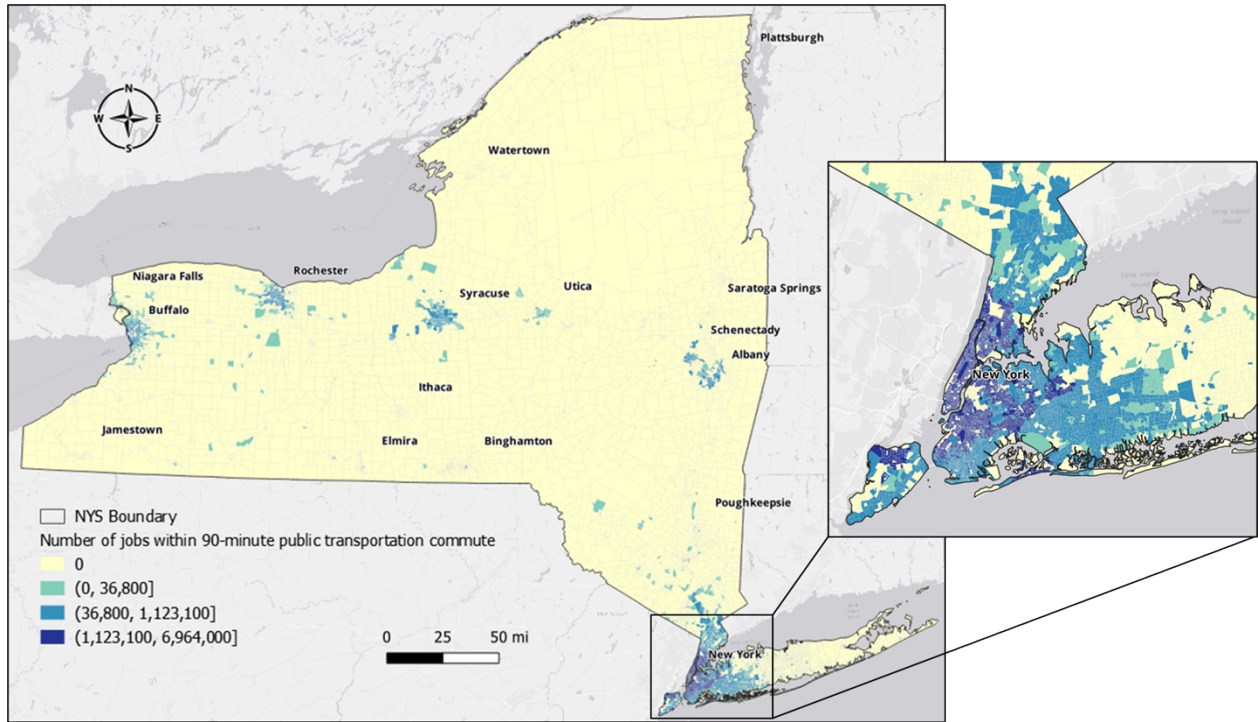


Figure 8. Number of jobs within a 90 min public transportation commute.

5. RESULTS

5.1 Model Fit

Table 4 presents the model fit results of the best-performing model based on backward model selection. The coefficients of the factors that influence the probability of being a night commuter of this model were all statistically significant at a 95% confidence level. For example, a coefficient of 0.736 indicates that individuals in the 25-39 age group have higher log-odds of being night commuters compared to the reference group (18-24). Longer distance to work was positively correlated with being a night commuter. The marginal effect of 0.076 for this variable indicated that individuals whose workplace is located 20 mi or more from their residence have a 7.6% higher likelihood of being a night commuter compared with those whose workplace is less than 20 mi away, holding all other variables constant. Additionally, higher household income and homeownership were also linked to a greater likelihood of being a night commuter. Furthermore, night commuters had lower likelihood to be female.

Table 4. Logistic regression results: Estimating the probability of being a night commuter

	Model estimation			Marginal effect	
	Coefficient t	Standard error	p-value	dy/dx	Standard error
Y = night commuters/not night commuters (N = 34,297 individuals)					
Constant	-2.146	0.071	0.000		
Long distance to work (i.e., over 20 mi) (Ref: Not long distance to work)	0.886	0.065	0.000	0.076	0.006
Female (Ref: Male)	-0.186	0.041	0.000	-0.016	0.004
Household income equal or below median (Ref: Higher than median)	-0.312	0.044	0.000	-0.027	0.004
Residence is rented (Ref: Owned)	-0.224	0.054	0.000	-0.019	0.005

Age between 25 and 39 (Ref: 18-24)	0.736	0.049	0.000	0.063	0.004
Road network density	0.662	0.223	0.003	0.057	0.019
Number of jobs within a 90 min public transportation commute	1.218	0.212	0.000	0.105	0.018
Chi-square likelihood-ratio test statistic	650.63 ($p<.001$)				
Log-likelihood	-8,479.8				
Akaike Information Criterion (AIC)	16,975.68				
Bayesian Information Criterion (BIC)	17,041.45				

In terms of land use and public transportation accessibility factors, night commuters generally resided in areas with higher road network density and better job accessibility via public transportation. This alignment with improved public transportation accessibility supports the earlier assumption that such factors contribute to the higher likelihood of night commuters using public transportation, as observed in Figure 4. A likelihood-ratio test was performed to compare the fitted model with the null model, which only includes an intercept and no predictors. The results showed a significant improvement in model fit, indicating that the inclusion of the predictors meaningfully contributes to explaining the variance in the outcome variable.

Table 5 presents the factors that influence public transportation use among night commuters. The coefficients were all statistically significant. Variables such as distance to work, gender, and the number of jobs within a 90 min public transportation commute all affected the likelihood of being a night commuter and using public transportation.

Table 5. Logistic regression results: estimating the probability of being a public transportation user

	Model estimation			Marginal effect	
	Coefficient	Standard error	p -value	dy/dx	Standard error
Y = existing/potential public transportation users (N = 3,312 night commuters)					
Constant	-3.375	0.186	0.000		
Long distance to work (i.e., over 20 mi) (Ref: Not long distance to work)	1.107	0.208	0.000	0.081	0.015
Female (Ref: Male)	0.260	0.145	0.072	0.019	0.011
Household owns 3+ vehicles (Ref: 0 vehicle)	-1.176	0.218	0.000	-0.086	0.016
Number of jobs within 0.5 mi of a fixed-guideway transit stop	0.233	0.033	0.000	0.017	0.002
Number of jobs within a 90 min public transportation commute	0.153	0.016	0.000	0.011	0.001
Chi-square likelihood-ratio test statistic	536.98 ($p<.001$)				
Log-likelihood	-671.08				
AIC	1,354.79				
BIC	1,390.08				

Although females had generally lower likelihood to be night commuters, those who did commute at night had higher likelihood to use public transportation. Additionally, individuals with longer commute

distances had higher likelihood to use public transportation. Households with three or more vehicles had lower likelihood to rely on public transportation compared to the reference group (zero vehicle).

Two specific metrics related to public transportation—namely, the number of jobs along fixed-guideway transit routes and the overall number of jobs accessible by public transportation —were positively correlated with the likelihood of being a public transportation user. Both factors contributed to a greater propensity to use public transportation.

5.2 Sensitivity Analysis

This sensitivity analysis was designed to examine how incremental increases in the number of accessible jobs (e.g., 100, 200, 300) would affect the likelihood of becoming a public transportation user. The results are shown in Figure 9.

The results indicated that the rate of increase in the likelihood of public transportation use diminishes after adding 1,000 new jobs for both metrics. On average, jobs within fixed-guideway transit areas had a greater impact on increasing the probability of becoming a public transportation user than jobs accessible within a 90 min public transportation commute. Specifically, when the number of jobs within fixed-guideway transit increased by 1,000, the probability of becoming a public transportation user exceeded 20%. A similar increase in jobs accessible within a 90 min transit commute resulted in a probability increase of just over 10%.

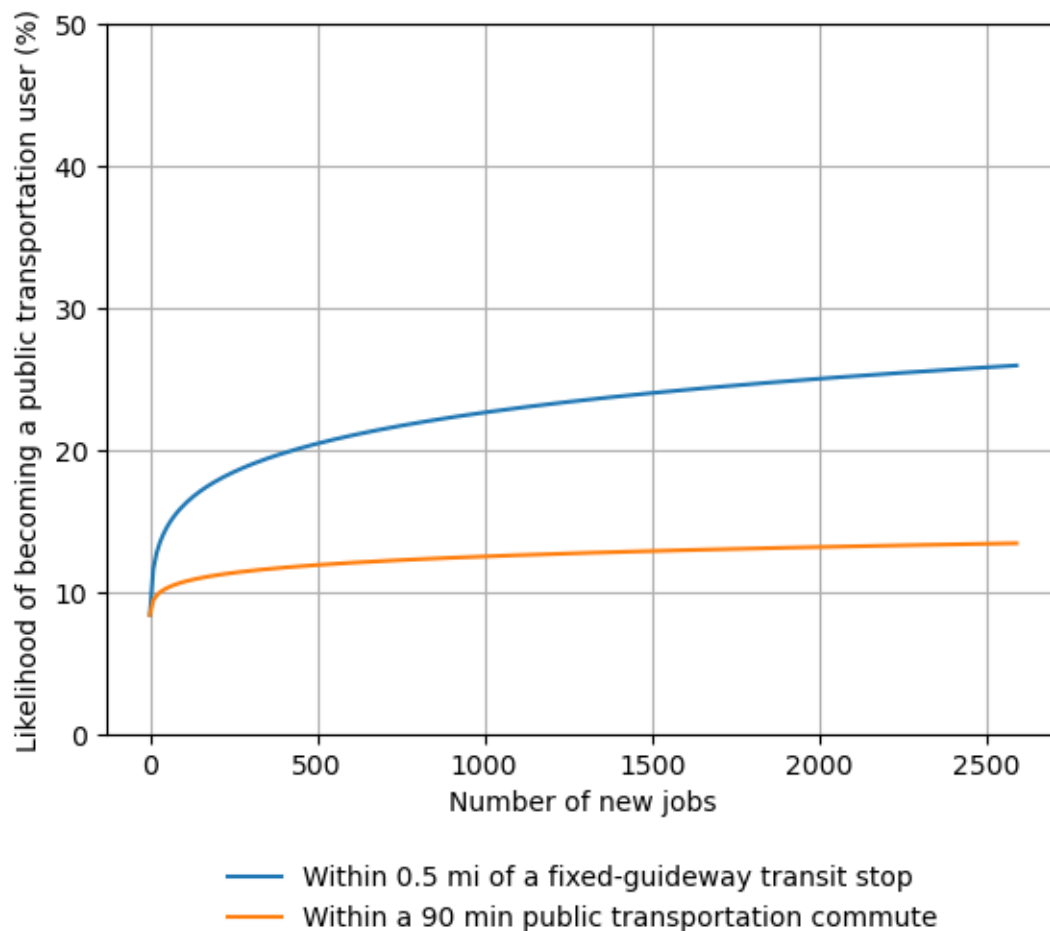


Figure 9. Likelihood of becoming a public transportation user given the number of new jobs.

The probability of becoming a public transportation user in areas where current non-users reside was estimated using the model presented in Table 4, and the results are shown in Figure 10. The shaded areas on the map represent regions currently without public transportation coverage, defined as areas lacking public transportation stops within 0.75 mi of the CBG population centroid.

Figure 10 shows regions that were more sensitive to an increase in jobs located within 0.5 mi of a fixed-guideway transit stop. These areas could benefit from targeted investments in job growth. The analysis showed a higher likelihood of increased public transportation use in NYC and other densely populated cities in NYS, such as Buffalo, Rochester, Syracuse, and Albany. Although most of these cities already had public transportation coverage, there were still regions, particularly in Elmira and Ithaca, where public transportation infrastructure may be insufficient.

Figure 11 shows regions with high sensitivity to increase in job availability within a 90 min public transportation commute. The highest probability was observed in the Manhattan region of NYC, where there was already a dense concentration of jobs accessible within a 90 min public transportation commute. Like in Figure 10, other regions with potential for increased public transportation use were in larger cities across NYS.

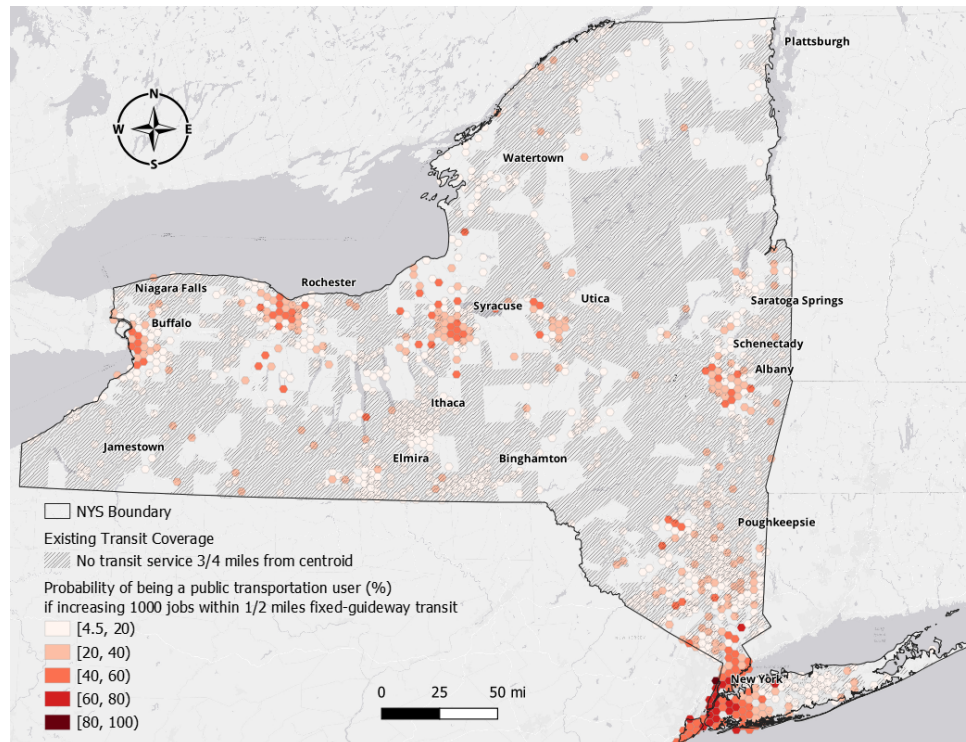


Figure 10. Probability of being a public transportation user if adding 1,000 jobs within 0.5 mi of a fixed-guideway transit stop.

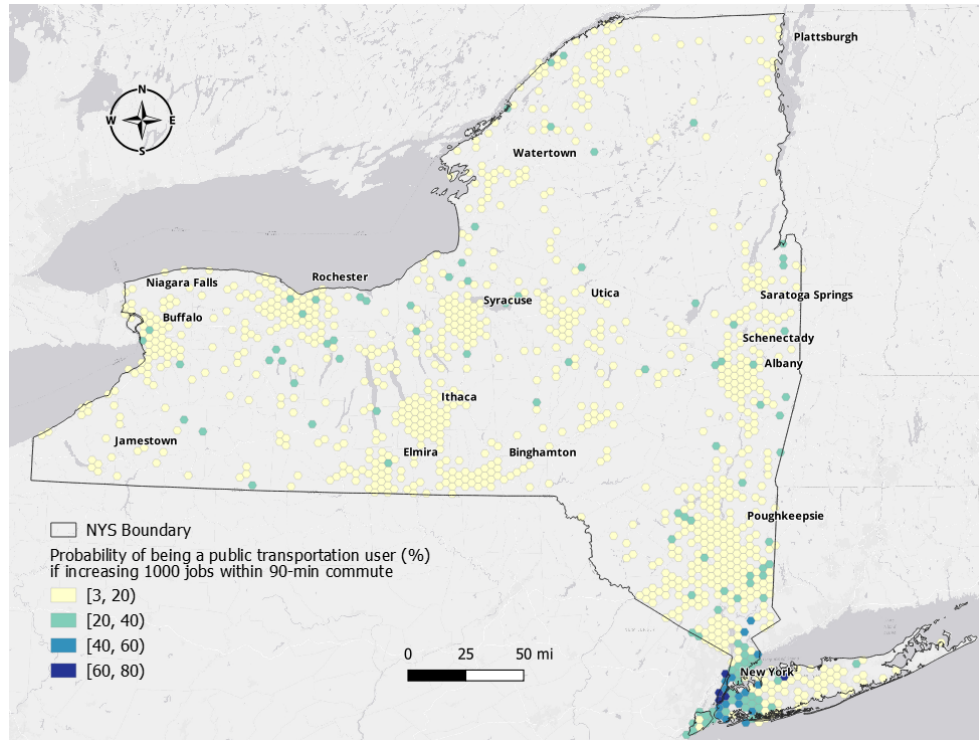


Figure 11. Probability of being a public transportation user if adding 1,000 jobs within a 90 min public transportation commute.

6. DISCUSSION

6.1 Characteristics of Night Commuters

Our analysis reveals several key characteristics of night commuters. First, a longer distance to work positively correlates with being a night commuter. This finding agrees with research on evening shift workers that night shift workers were slightly more likely to face commutes lasting over an hour (42, 43). Moreover, our study indicated that night commuters are less likely to be female and more likely to be between 25 and 39 years old. This demographic insight has also been found in labor market studies showing that younger adults and males without children are more inclined to work non-standard hours (44). Higher household income and homeownership were associated with night commuting. This implies that individuals in higher-paying jobs, which may require flexible or extended hours, are more likely to work night shifts. The flexibility provided by night shifts may be necessary for meeting the demands of these roles. However, other studies found that shift workers are more often low-income and non-white (42, 45), a pattern not observed in our study. This discrepancy may be because of differences in sample composition; those studies included a higher proportion of immigrants and low-income individuals. Additionally, the NHTS lacks detailed occupational descriptions, which limits our ability to more fully examine these results.

Night commuters were more likely to reside in areas with higher road network density and better job accessibility via public transportation. This finding indicates that urban planning and the availability of efficient public transportation could provide crucial support for night commuting.

6.2 Characteristics of Public Transportation Users

Our study extends previous research on public transportation service quality and job accessibility influencing public transportation use (46). Specifically, we investigated which public transportation-related attributes attract night commuters. Although females are generally less likely to be night

1 commuters, those who are night commuters are more likely to use public transportation. This pattern
2 might be attributed to several factors. One study suggested that women are less likely to drive at night
3 because of concerns such as vision issues and general discomfort with nighttime driving (47).
4 Additionally, research has shown that women often perceive public transportation as a safer option than
5 walking or waiting for ride-hailing services late at night (5). The well-lit environment of public
6 transportation stations and the presence of other passengers can provide a sense of security, making public
7 transportation a more appealing choice for women traveling at night. However, this perspective is not
8 universally held. Some studies also suggested that women perceived public transportation as less safe in
9 the evening (48, 49). This gender-specific pattern suggests that services and safety measures must be
10 tailored to address the unique needs and concerns of female night commuters.

11 Our study also indicates that the increased likelihood of longer commute distances led to higher public
12 transportation use. The relationship between longer commutes and night work may also be linked to
13 safety concerns, especially for those returning home in the early morning hours. Driving after an hour-
14 long shift at 3 a.m. while fatigued could be risky, and public transportation may offer a less stressful
15 alternative for longer distances. Night shift workers often face irregular sleep schedules, and using public
16 transportation for longer commutes could provide an opportunity to rest and relax (50). This restorative
17 use of time contrasts with the heightened stress and safety concerns of driving long distances late at night
18 or early in the morning. While much of the literature on safety focuses on the sense of security and fear of
19 crime—particularly among women (21, 51)—the impact of safety on night commuters' choice between
20 public transportation and driving has not been adequately studied. Our findings suggest that public
21 transportation may play a crucial role in supporting the safety of night commuters by reducing crash risks
22 associated with driver fatigue. Future research could explore not only how public transportation provides
23 security but also how the interior of buses can be made more comfortable to facilitate rest for riders.

24 Household vehicle ownership significantly influences public transportation use patterns among night
25 commuters. Our findings reveal that households with three or more vehicles exhibit lower public
26 transportation use, consistent with the literature on the general population indicating that higher vehicle
27 ownership correlates with a preference for using private vehicles because of their convenience and
28 flexibility (52). This finding might also present an opportunity for public transportation agencies to
29 develop strategies that can attract car owners to public transportation, such as park-and-ride facilities or
30 incentives for using public transportation during off-peak hours.

31 Our study identified two key public transportation-related metrics that are correlated with a higher
32 likelihood of being a public transportation user: the number of jobs along fixed-guideway transit routes
33 and the overall number of jobs accessible by public transportation. Both metrics positively contributed to
34 a greater inclination to use public transportation, emphasizing the importance of job accessibility in
35 promoting public transportation use. Therefore, transit-oriented development strategies should focus not
36 only on residential areas but also on creating job centers along public transportation corridors. Fixed-
37 guideway transit systems, such as subways and dedicated bus lanes, could offer reliable and efficient
38 transportation options crucial for night commuters. These metrics also helped explain the high density of
39 night commuters observed in NYC, where the greater road network density and job accessibility via
40 transit facilitate both night commuting and public transportation use.

41 However, we did not find a significant influence of public transportation frequency during evening peak
42 hours on public transportation use among night commuters. This variable may not be well-suited for
43 modeling public transportation use among night commuters, whose travel patterns may differ
44 significantly from daytime commuters. Future research could consider examining late evening public
45 transportation frequency as a potential predictor variable, which may more accurately reflect the service
46 availability to night commuters.

6.3 Potential of Public Transportation Use among Night Commuters

In urban areas with substantial public transportation infrastructure, we observed that additional job opportunities could further enhance public transportation use. This finding aligns with some principles of transit-oriented development that increasing job density around public transportation hubs could promote sustainable urban mobility (53). Regions that currently lack adequate public transportation coverage, such as Elmira and Ithaca, also exhibited potential for increased public transportation use with targeted investments in public transportation infrastructure. This observation likely stems from demographic and socioeconomic characteristics that align with typical public transportation users, such as lower-income households or a higher proportion of students and young professionals who are more inclined to use public transportation if available.

This study reveals a high sensitivity to job increases within a 90 min public transportation commute in certain regions, with Manhattan showing the highest probability. Despite the area's existing high job concentration, some individuals may still face relatively low job accessibility via transit. However, the comprehensive public transportation network in these areas means that even marginal increases in job availability can significantly influence mode choice.

6.4 Limitations

Individuals' perceptions and attitudes were not included in this study. Consequently, the estimates from the sensitivity analysis represent the rough approximations of the direction and magnitude of changes in public transportation use. These estimates assume that the impact of other factors is minimal and consistent across regions. For more detailed insights, incorporating a wider range of explanatory factors into the models would be beneficial.

Since the NHTS does not collect data on occupation or specific shift types, this study includes individuals with both standard evening shifts (starting as late as 6 p.m.) and those working overtime. The sample may also include both low- and high-wage workers from different job sectors. These groups have varying socioeconomic statuses and reasons for their mode choices. We acknowledge this limitation and plan to address it using other datasets in future research.

Additionally, while the literature highlights the role of attitudes and perceptions in public transportation use, our study focused on land use and accessibility factors to understand their specific impact. The public transportation frequency data in the SLD does not differentiate nighttime schedules, and we did not explore other metrics that assess nighttime service levels. Future studies could incorporate data from regional transit agencies, such as bus schedules in General Transit Feed Specification format, to better evaluate the impact of public transportation on night commuters.

While this study provides a robust framework for analyzing night commuters and the factors influencing their public transportation use, the transferability of these results to other regions should be considered with caution. The unique socio-economic, land use, and transportation characteristics of NYS, particularly in urban areas like NYC, may limit the applicability of these findings to regions with different urban densities, transit infrastructures, or job distributions. Future research could replicate this framework in diverse settings to validate its broader applicability and identify region-specific factors influencing night commuters' use of public transportation.

7. CONCLUSIONS

In this study, we applied logistic regression to understand the characteristics of night commuters and the land use and public transportation accessibility factors that contribute to their use of public transportation. We analyzed the 2017 NHTS survey data, which includes 3,312 night commuters, to examine nighttime mobility in NYS. The major findings are as follows:

- Night commuters were more likely to be male, aged 25-39, and have higher household incomes and homeownership. These commuters tended to live in areas with higher road network density and better public transportation accessibility.
- Although females are generally less likely to be night commuters, those who are night commuters are more likely to use public transportation, potentially due to discomfort with nighttime driving.
- Longer commutes were associated with higher public transportation use, likely as it provided a safer, less stressful alternative for fatigued workers.

Public transportation agencies can use the framework provided in this study to identify regions that might benefit more from increased services for night commuters. Specifically, the application of this method in NYS demonstrated the following:

- The number of jobs along fixed-guideway transit routes and overall job accessibility via public transportation positively influence the likelihood of using public transportation.
- Among areas with no existing public transportation users, an increase of 1,000 jobs within fixed-guideway transit routes increase the probability of public transportation use by more than 20%. For example, regions with limited public transportation, such as Elmira and Ithaca, could see increased use with targeted investments. Similarly, an increase of 1,000 jobs accessible within a 90 min public transportation commute increases the probability by just over 10%. For example, job density increases within a 90-minute public transportation commute can significantly influence mode choice, especially in areas like Manhattan, where even small job access improvements can impact public transportation use.

Public transportation agencies and urban planners could use findings from this study to identify areas for improvement and to estimate potential increases in transit use. The modeling approach helps to create more inclusive, efficient, and attractive public transportation systems that meet diverse needs. By enhancing the job accessibility of public transportation, agencies can benefit current night workers and enable more individuals to pursue nighttime employment opportunities, especially those who do not have reliable access to personal vehicles.

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